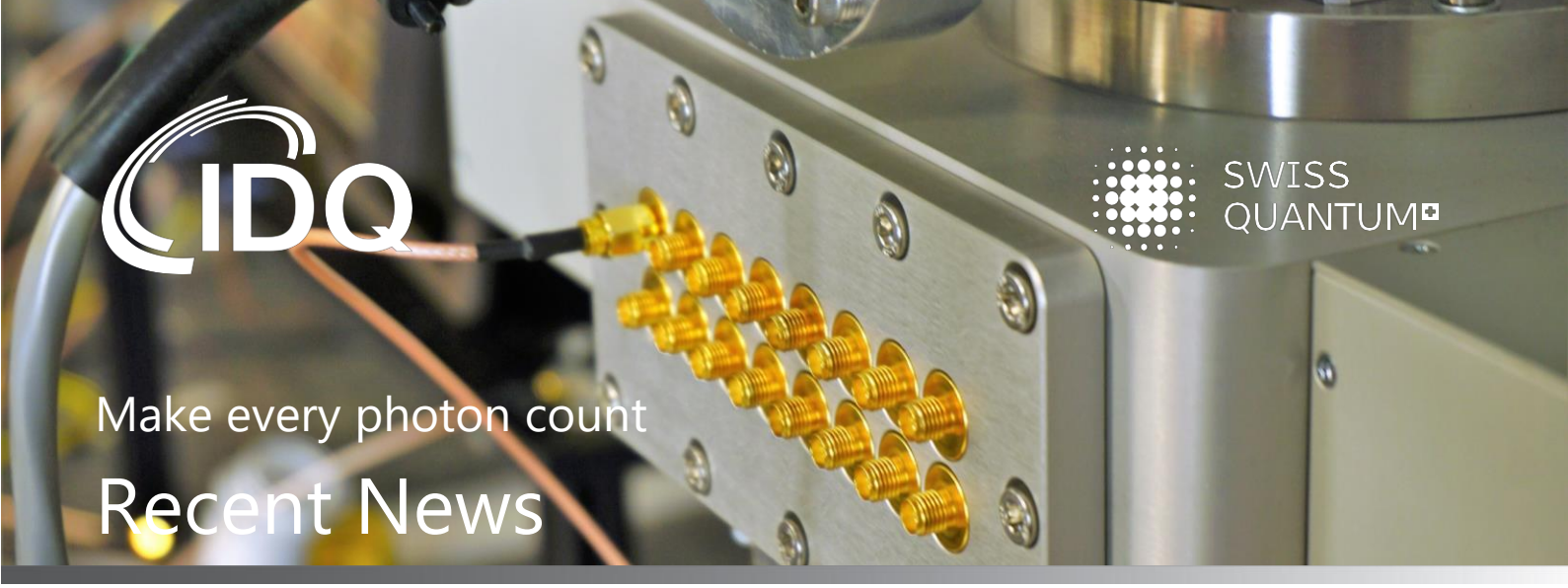


Make every photon count

Recent News

2022 Photonic Sensing Research Digest

Here we sample some of the exciting and impactful photonic sensing work in 2022 so far, from researchers around the world

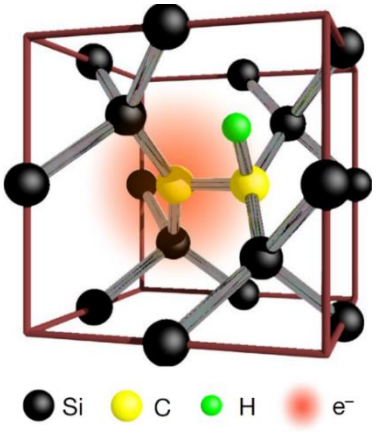
Linking silicon qubits to photonic qubits 1
Increasing laser power without sacrificing size 2
Advances in fibre-integrated quantum memories 2
Single-photon control with quantum memories 2
Simultaneous pairwise QKD 3
Exploiting quantum asymmetry 3
Scalable and high-capacity quantum light sources 3

Linking silicon qubits to photonic qubits

At the heart of the Quantum Internet lies the ability to interconvert flying photonic qubits (fast and far-reaching) and matter-based qubits (stationary and useful). Recently, researchers at Simon Fraser University were able to optically address the spin-qubits of their silicon-based devices with photons in the telecom O band.

This is a foundational opportunity to develop some of the key infrastructure components for future scalable and high-performance quantum telecommunication infrastructure.

From IDQ, this work employed ID281 Superconducting Nanowire Series detectors, ID230 Infrared Single-Photon Detector, and a Time Controller Series device.

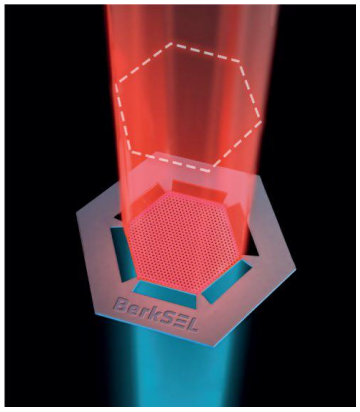


From DB Higginbottom et al. *[1]*

[1] **Optical observation of single spins in silicon**
DB Higginbottom, ATK Kurkjian, C Chartrand, M Kazemi, NA Brunelle, ER MacQuarrie, JR Klein, NR Lee-Hone, J Stacho, M Ruether, C Bowness, L Bergeron, A DeAbreu, SR Harrigan, J Kanaganayagam, DW Marsden, TS Richards, LA Stott, S Roorda, KJ Morse, MLW Thewalt and S Simmons
Nature 607, no. 7918 (2022): 266-270, published online 13 July 2022

Find out more **Read the paper**

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1227 Carouge/Geneva
Switzerland
info@idquantique.com
www.idquantique.com



From R Contractor et al. *[2]*

Increasing laser power without sacrificing size

Scaling up the power of a laser isn't generally feasible without harming the coherence of the laser light, where desired single modes are overcome by higher-order transverse modes as the cavity size increases. Researchers at UC Berkeley and the Berkeley Lab have been able to demonstrate a novel laser cavity design, where its unconventional behaviour is exploited to allow larger and larger lasers to operate. This work opens a range of applications and avenues of investigation in light-matter interaction and cavity quantum electrodynamics.

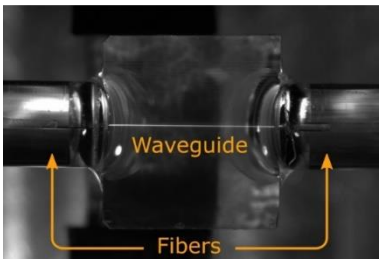
From IDQ, this work employed ID281 Superconducting Nanowire Series detector to detect the micro-photoluminescence collected in characterizing the Dirac Laser.

[2] **Scalable single-mode surface emitting laser via open-Dirac singularities**
R Contractor, W Noh, W Redjem, W Qarony, E Martin, S Dhuey, A Schwartzberg, and B Kanté
Nature 608, 692–698 (2022), published online 29 June 2022

Find out more **Read the paper**

Advances in fibre-integrated quantum memories

Photons can be hard to steer without destroying their useful quantum coherence properties, so the investigation of quantum memories for "buffering" photonic states is important for future quantum computing architectures. In this work, researchers at ICFO, IFN-CNR and Heriot-Watt University demonstrate entanglement between a fibre-integrated quantum memory and a telecommunications-wavelength photon, an important stepping stone for long-distance quantum communication.

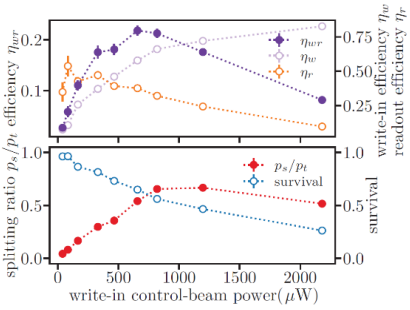


From JV Rakonjac et al. *[3]*

From IDQ, ID281 Superconducting Nanowire Series detectors were used to herald the light-matter entangled state.

[3] **Storage and analysis of light-matter entanglement in a fiber-integrated system**
JV Rakonjac, G Corrielli, D Lago-Rivera, A Seri, M Mazzera, S Grandi, R Osellame, and H de Riedmatten
arXiv:2201.03361 (2022), published online 8 July 2022

Find out more **Read the paper**



From L Heller et al. *[4]*

Single-photon control with quantum memories

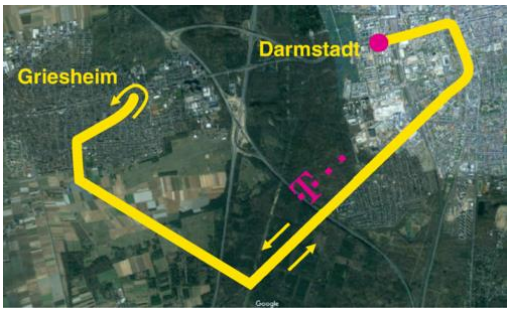
While fibre-integration is important for quantum memories to be useful, they must also have sufficiently precise and efficient storage and reading out of their photonic inputs. Here, researchers at ICFO demonstrate the storage and retrieval of single photons in a low-noise Raman quantum memory.

This work used ID281 Superconducting Nanowire Series detectors to alternately read out or perform autocorrelation measurements on stored photons.

[4] **Raman Storage of Quasideterministic Single Photons Generated by Rydberg Collective Excitations in a Low-Noise Quantum Memory**
L Heller, J Lowinski, K Theophilo, A Padrón-Brito, and H de Riedmatten
Physical Review Applied 18, no. 2 (2022): 024036, published online 12 August 2022

Read the paper

Simultaneous pairwise QKD



From E Fitzke et al. *[5]*

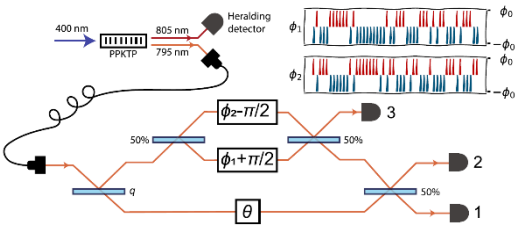
Researchers at TU Darmstadt have demonstrated a scalable star-shaped QKD network, with an entanglement-based protocol involving simultaneous key exchanges between four participants.

This work employs ID220 Single Photon Counters (precursors to the ID Qube NIR Series), and a Time Controller Series device, with a discussion on the distance and key-rate improvements that would have been experienced with the use of SNSPDs such as the ID281 Superconducting Nanowire Series detectors.

[5] **Scalable Network for Simultaneous Pairwise Quantum Key Distribution via Entanglement-Based Time-Bin Coding**
E Fitzke, L Bialowons, T Dolejsky, M Tippmann, O Nikiforov, T Walther, F Wissel, and M Gunkel
PRX Quantum 3, no. 2 (2022): 020341, published online 24 May 2022

Read the paper

Exploiting quantum asymmetry



From F Albarelli et al. *[6]*

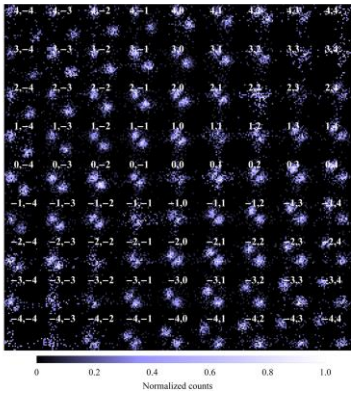
Like quantum coherence, quantum asymmetry is a physical resource that can increase the sensitivity of measuring equipment. In this work, researchers at the University of Warsaw demonstrate an interferometric experiment that sees an increase in sensitivity despite a decrease in correlation between the reference signal fluctuations.

This work employed ID281 Superconducting Nanowire Series detectors from IDQ for all single-photon detections.

[6] **Quantum Asymmetry and Noisy Multimode Interferometry**
F Albarelli, M Mazelanik, M Lipka, A Streltsov, M Parniak, and R Demkowicz-Dobrzański
Physical Review Letters 128, no. 24 (2022): 240504, published online 17 June 2022

Read the paper

Scalable and high-capacity quantum light sources



From S Merkouche et al. *[7]*

As a resource for quantum computation and communication, the photon pairs generated with time-frequency entanglement are relatively easy to generate, and have a high capacity for encoding information. Researchers at the University of Oregon demonstrated this in a versatile and scalable entanglement swapping experiment, which required no special source engineering.

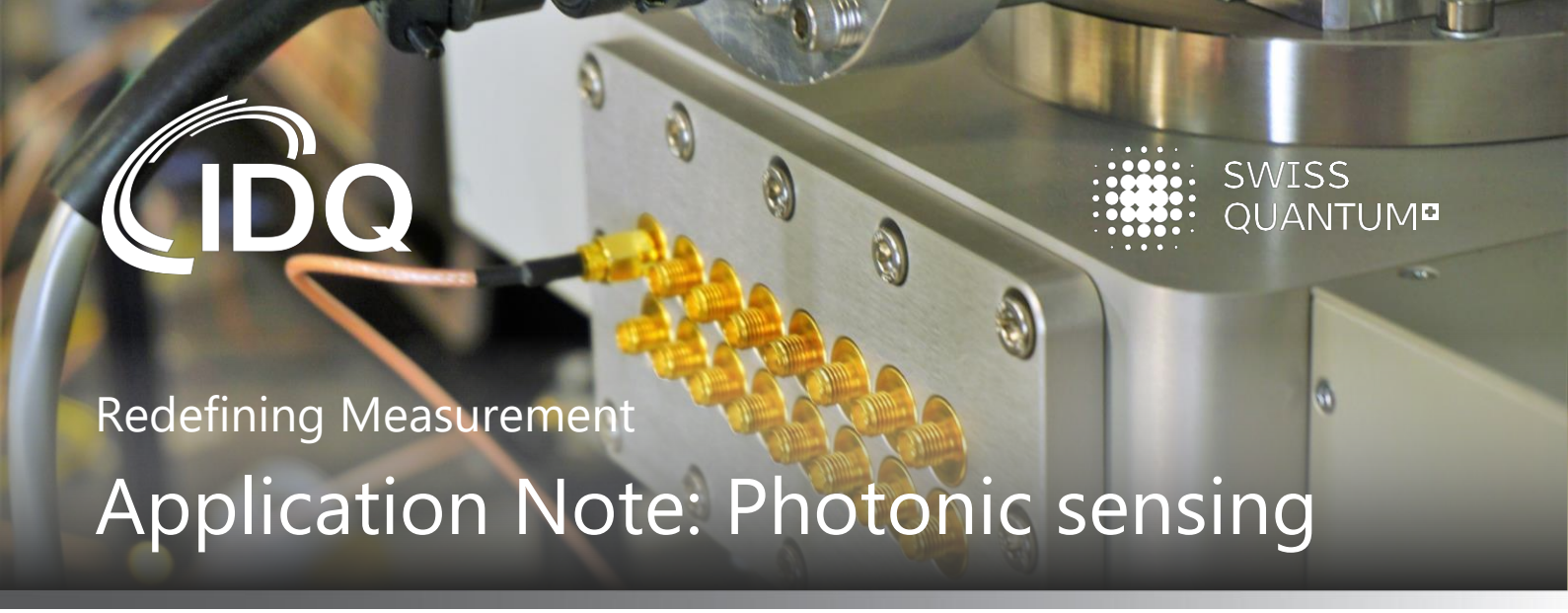
From IDQ, this work employed ID281 Superconducting Nanowire Series detectors, and a Time Controller Series device, to detect and record all single-photon events.

[7] **Spectrally resolved four-photon interference of time-frequency-entangled photons**
S Merkouche, V Thiel, and BJ Smith
Physical Review A 105, no. 2 (2022): 023708, published online 11 February 2022

Read the paper

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Redefining Measurement

Application Note: Photonic sensing

High-speed single-photon detection

Taking single-photon detection rates to new highs, with the ID281 parallel-pixel superconducting detectors.

Detecting single photons is hard. Not because they are strange wonders of quantum mechanics, as you might think, but because there are simply too many of them. One has a similar problem counting the number of drops of water in the ocean.

It is a relatively simple matter[1] to turn a single particle of light into a machine-readable signal. Photo-multiplier tubes (PMTs) and single-photon avalanche diodes (SPADs) are examples of detectors that rapidly turn a single photon into a large number of electrons, much like the clicks of a Geiger-Mueller tube detecting particles of ionising radiation. The state of the single-photon detection art, however, lies with superconducting nanowire detectors.

In this application note, we look at the challenge of high-speed single-photon counting, including the photon pile-up effect and a novel detection approach designed to circumvent this, allowing you to detect photons faster than ever before, above 200 MHz and beyond.

In good news, the parallel-pixel system design (patent pending) presented here, with detector recovery times shorter than 10 ns, is commercially available from IDQ today.

MEET THE PRODUCTS



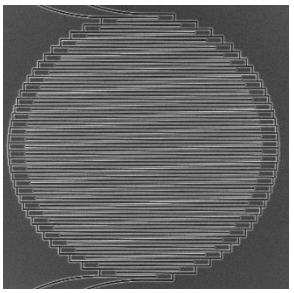
ID281 Superconducting Nanowire Parallel-Pixel model

References

[1] **Operation of parallel SNSPDs at high detection rates**
Matthieu Perrenoud, Misael Caloz, Emna Amri, Claire Autebert, Christian Schönenberger, Hugo Zbinden, and Félix Bussières
Supercond. Sci. Technol. 34 (2021) 024002

[1] Read: not actually simple, but we can do it with off-the-shelf technology.

The state of the art: SNSPDs



The present-day standard for single-photon detection is the superconducting nanowire single-photon detector (SNSPD). In this device, a single meandering line of superconducting material is held close to its superconducting threshold, where a single photon will create a localised but measurable resistance spike in the nanowire. This method provides high efficiency, precise timing and low noise, which allows SNSPDs a much wider range of photonic experiments than their SPAD counterparts:

- **Detection is near-ideally efficient**. Close to 100% of individual photons are detected, compared to about 40% efficiency for high-performance telecom-wavelength SPADs.
- **Timing is highly precise and robust**. Pulses are repeatably measured within tens of picoseconds of each other, compared to the hundreds of picoseconds of high-performance InGaAs SPADs. This timing jitter is wavelength dependent: higher-energy (shorter-wavelength) photons will create a steeper response from the detector, and thus provide a greater level of timing precision. This timing jitter is usually the dominating factor in the time resolution of a detector system.
- **Noise is ultra-low**. Electrical noise in the detector is effectively zero, and by operating at cryogenic temperatures, the number of black-body photons that may cause a false detection event can be less than one photon per second at visible wavelengths, but increases at higher wavelengths.

The Problem

Detectors can only count so many events before they begin to saturate. The maximum count rate a system can measure efficiently is dictated by the detector recovery time: each time a detector is triggered, it requires a short but finite time to reset its electrical configuration. Photons hitting the detector during this recovery time are not registered as detected events.

The result is known as the photon pile-up effect. When the incident flux of photons increases, the capacity to detect newly arriving photons decreases, and the overall system detection efficiency decreases (see Figure 1). The shorter the recovery time, the greater the maximum number of photons that can be detected well. For InGaAs SPADs, this recovery time is typically hundreds of nanoseconds, giving a maximum count rate of several MHz for the best systems. For SNSPDs, this time is shorter, typically recovering in a few tens of nanoseconds. As such, SNSPDs can have a maximum count rate of several tens of MHz before the detectors saturate.

Detecting fifty million-or-so photons per second is respectable, but can we do better?

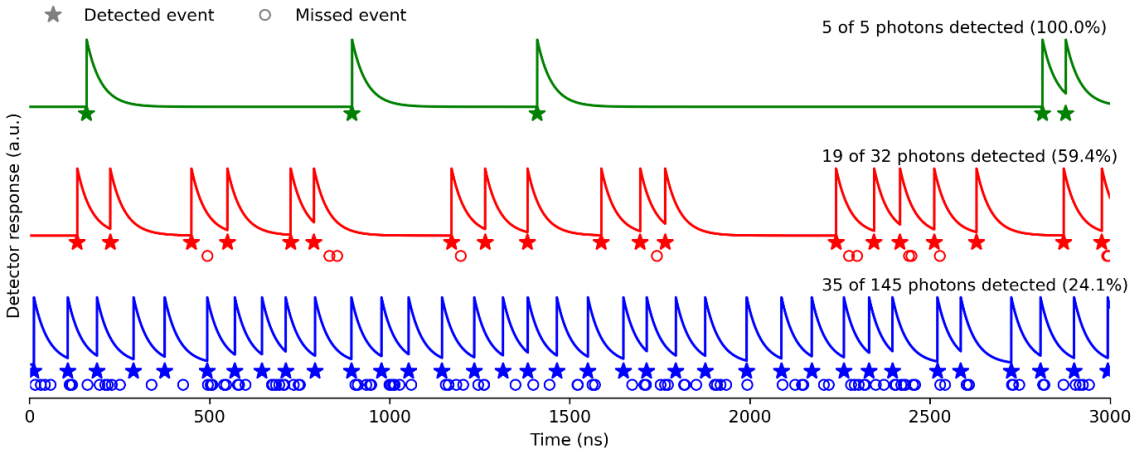


Figure 1: Representation of the photon pile-up effect, demonstrated for detector inputs of approximately 2 MHz (top, green), 10 MHz (middle, red) and 50 MHz (bottom, blue). Detected photons are marked with a star, while missed photons that arrive at the detector too soon after a detection event are marked with an empty circle. The lower the single-photon rate, the greater the proportion of photons that are detected, and therefore the higher the system detection efficiency.

The Solution

We can indeed do better. A neat solution to the photon pile-up effect is a parallel-pixel detector design, as recently demonstrated by Matthieu Perrenoud *et al.* [1], in a collaboration between IDQ and the group of Hugo Zbinden at the University of Geneva. Here, an array of superconducting detectors are connected electrically in parallel, to a single readout circuit (see Figure 2). While each individual pixel is still limited by a required recovery time, by distributing the signal to be measured across the array, the arriving photons are now much less likely to pile up at one detector, and considerably higher single-photon detection rates can be achieved.

All this is still through a single-channel readout, which is a must-have for scaling this high-speed approach in an SNSPD system. If there was a separate readout channel for each pixel, the supporting electronics would quickly become very complex, and in the cryogenic environment of the SNSPD system, the added thermal load would degrade the stability and noise performance of the detector. This parallel pixel design, as such, is an elegantly scalable approach towards GHz-rate single-photon detection.

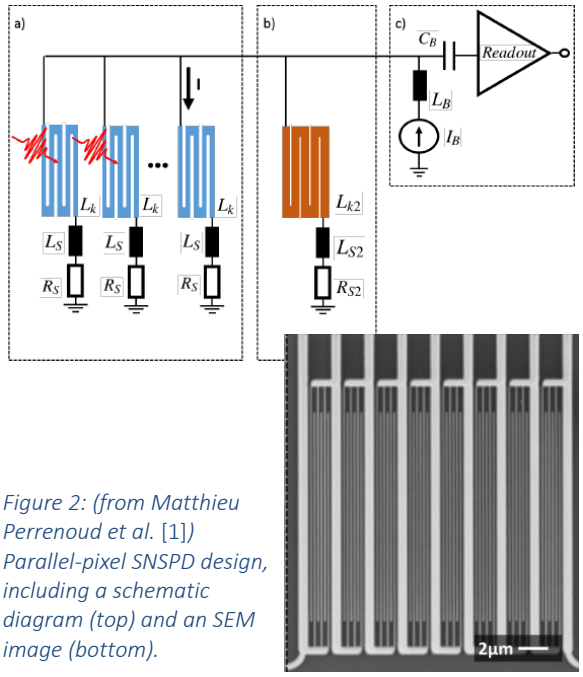


Figure 2: (from Matthieu Perrenoud et al. [1]) Parallel-pixel SNSPD design, including a schematic diagram (top) and an SEM image (bottom).

The Result

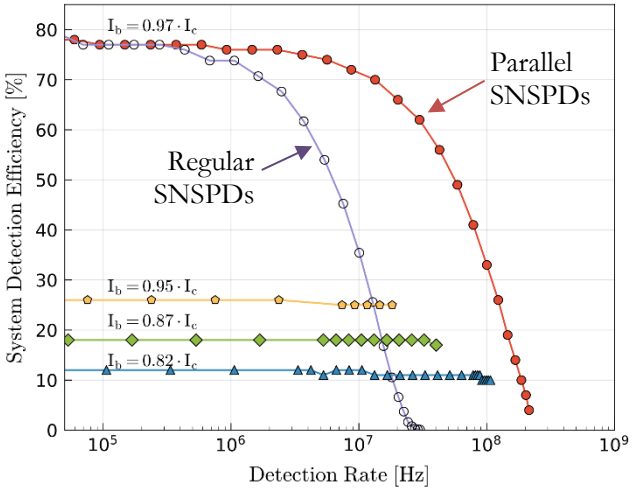


Figure 3: (from Matthieu Perrenoud et al. [1]) Changes in system detection efficiency with the measured detection rate, for regular SNSPDs (lower/left-hand curve, violet) and parallel-pixel SNSPDs (upper/right-hand curve, red).

The performance of the University of Geneva's device exceeded expectations. As shown in Figure 3, the low-rate detection efficiency for the parallel SNSPD design was as strong as a regular single-pixel SNSPD, comfortably above 75% efficiency. At high count rates, however, the parallel design comes into its own.

At a detection rate of 10 million photons/s, the regular SNSPDs' detection efficiency had dropped significantly, and was essentially zero beyond 20 million photons/s. In contrast, the parallel-pixel SNSPDs were still operating at close to peak performance.

Increasing the detection rate further, the parallel SNSPDs can count over 100 million photons/s while still above 30% system efficiency. Impressively, the parallel-pixel detector still counts 200 million photons/s with a system efficiency of 6.6%.

This order-of-magnitude improvement is a valuable addition to an already high-performance instrument: the ultra-low noise and superb timing precision of current SNSPD technology, with the high count-rates needed for future-proof TCSPC. From the quantum optics lab, to long-haul quantum communication, to high-precision high-sensitivity microscopy, LiDAR and photonic sensing applications, we can't wait to see how your research will benefit.

To discuss your high-speed single-photon applications with us, or to enquire about the ID281 Superconducting Nanowire Parallel-Pixel model, email us at info@idquantique.com, or visit us at www.idquantique.com/quantum-sensing today.

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Redefining Measurement

Use Case: Quantum Optics

Towards a Quantum Internet

Customer: ICFO Institute of Photonic Sciences

Research Field: Quantum Optics

Country: Spain

Customer Need

In heralded-entanglement experiments, ICFO needed to collect photons as efficiently with as little background noise as possible

Solution

ID281 SNSPDs with > 80% Quantum Efficiency and < 14 Hz Dark Count Rate at 1450 nm

Results

Successful entanglement of remote quantum memories and entanglement between a telecom-wavelength photon and an on-demand multimode quantum memory

Context

The ability to distribute entanglement across a quantum network is of paramount importance in a future-proof quantum internet. In building quantum communication channels, it would help to extend the channels' range through quantum repeater technology. Later, it will allow for distributed quantum computing, sharing arbitrary quantum states across a range of remote nodes, giving access to a richer repertoire of quantum computing protocols.

Three key capabilities are needed for this kind of remote entanglement distribution:

1. Generating quantum-entangled states,
2. Storing quantum states for controllable lengths of time with high fidelity, and
3. Interconverting quantum states between flying qubits (typically telecom-wavelength photons) and stationary qubits (typically massive particles).

Samuele Grandi, Dario Lago-Rivera, Jelena V. Rakonjac and Alessandro Seri, in the group of Hugues de Riedmatten at ICFO in Barcelona, set themselves the task of marrying these three capabilities together, and have recently demonstrated just that, in 'Telecom-heralded entanglement between multimode solid-state quantum memories' (Nature 2021) [1] and 'Entanglement between a telecom photon and an on-demand multimode solid-state quantum memory' (arXiv 2021) [2].

Customer Need

The work at ICFO centres around the heralded quantum entanglement of photon pairs, generated through spontaneous parametric down conversion (SPDC) and the subsequent storage of the entangled photons in Quantum Memories with atomic frequency comb (AFC) and spin-wave (SW) storage.

In the ICFO group's entanglement source, a pair of strongly correlated photons are generated in a frequency-mixing process such as SPDC. Sending each photon of the pair through different arms of the experiment, the arrival of the 'idler' photon at one detector tells the user when and where they can find a single 'signal' photon. In this way, observing heralded single photons requires a coincidence measurement of two photons.

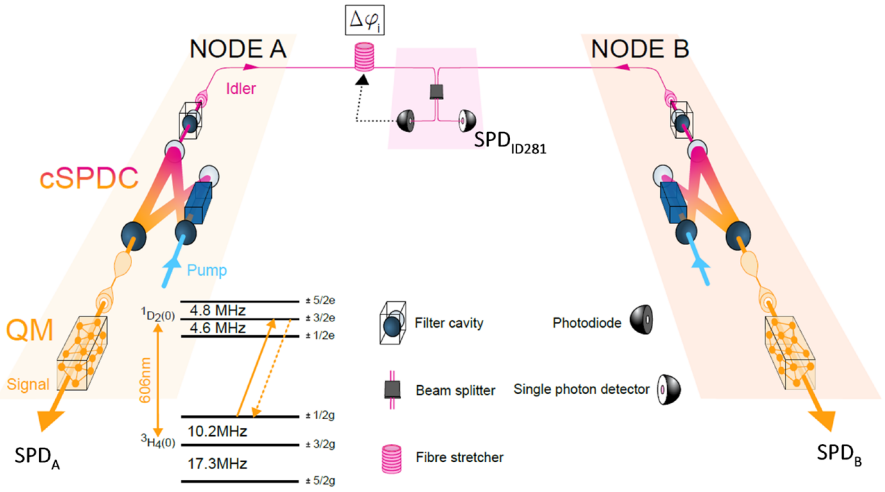


Figure 1: Configuration of the two nodes of the experiment, taken from D. Lago-Rivera, S. Grandi et al. (2021) [1]. Each node has its own SPDC source and quantum memory. The detection of a 1436 nm idler photon at the central ID281 SNSPD indicates the presence of a single 606 nm signal photon in the two nodes, and the subsequent entanglement of the two quantum memories.

The ICFO group's experiments expand on this: two photon pair sources are used to herald entanglement between two remote quantum nodes (or between one node and one signal photon). In their arrangement (see Figure 1), they only have one signal photon in the system at a time. However, to be sure they're not experiencing unwanted two-photon events, they need to measure the rate of triple-coincidences between detectors SPD_{ID281}, SPD_A and SPD_B in their experiment. Expanding their experiment into entanglement swapping, they may later want to measure quadruple-photon coincidences.

Solution

Performing these coincidence measurements requires time-correlated single-photon counting (TCSPC). Since the photon detection statistics in TCSPC follow a Poissonian distribution, detector efficiency and the detector dark count rate were of primary importance. An improvement in the efficiency would see a corresponding decrease in the time needed to build equivalent statistics by between a power of two and three. A decrease in the dark count rate would have a similar improvement on the quality of the measured statistics.

As such, ID Quantique provided ICFO with a set of ID281 superconducting nanowire single-photon detectors (SNSPDs), SPD_{ID281} in the experimental setup in Figure 1. These detectors have greater than 80% efficiency at the wavelength of interest (c. 1450 nm for the idler photon detection) and < 14 Hz dark count rate.

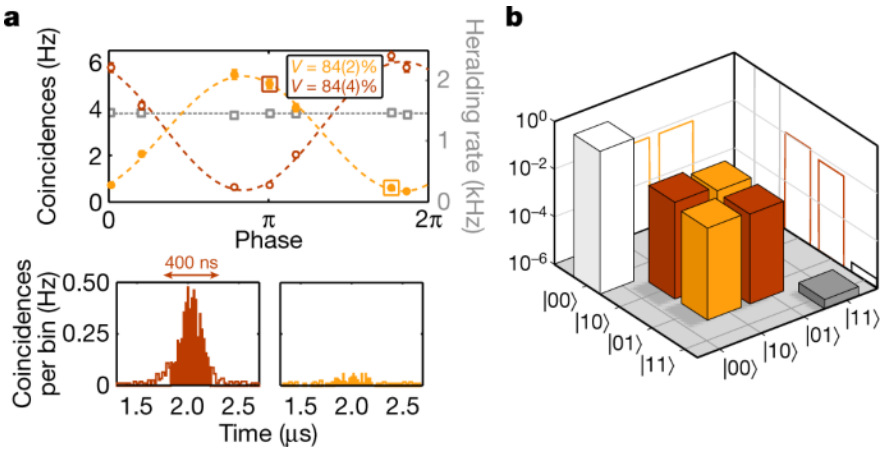


Figure 2: Measurements of the recombined node A and node B signal photons, after a storage time of 2 μs, taken from D. Lago-Rivera, S. Grandi et al. (2021) [1]. Coincidences are measured between detection events at the ID281 SNSPD and one of two detectors after the combined node A & B signal.

Results

ICFO demonstrated successful entanglement of remote quantum memories [1] and, for the first time, entanglement between a telecom-wavelength photon and an on-demand multimode solid-state quantum memory [2], all in a practical timeframe, laying the groundwork for a future quantum repeater.

> *The high efficiency and low dark counts of the new ID281 detectors reflected directly into an improved signal-to-noise ratio and in dramatic reduction in the required experimental time, allowing for short and consistent measurements and relaxing the need for long-term stabilisation.*
>
> Samuele Grandi
> ICFO Institute of Photonic Sciences

They achieved matter-matter entanglement between two quantum memories, with 2 µs storage times at a heralded entanglement rate of 1.43 kHz. This marks a 40-fold improvement on the current record between quantum memories. Even at longer storage times, up to 25 µs, they saw concurrences more than five standard deviations above zero, indicating continued entanglement, which would allow entanglement distribution across 5 km of optical fibre.

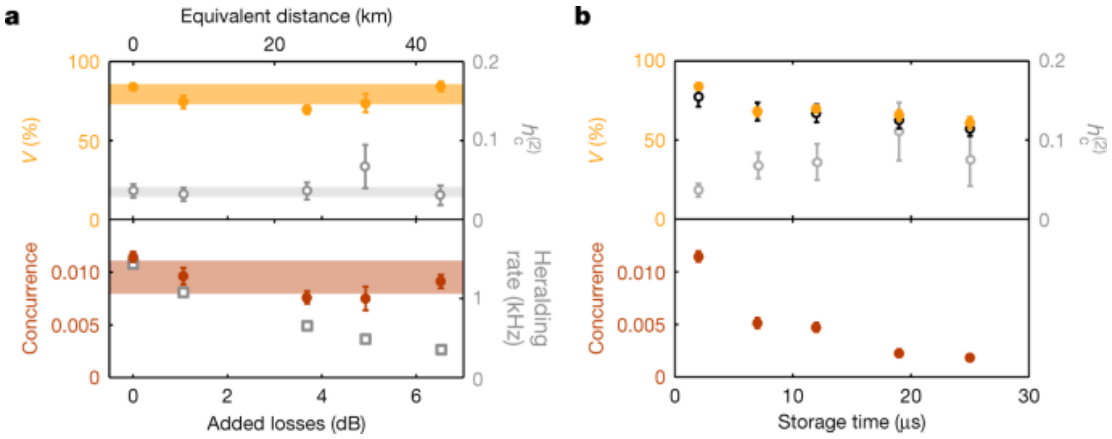


Figure 3: Measurement of the overall quantum state's visibility and concurrence for (a) artificially added channel losses, simulating extended distances, and (b) increases in the QM storage time up to 25 µs (taken from D. Lago-Rivera, S. Grandi et al. (2021) [1]).

This research demonstrates some of the key building blocks needed to build a scalable quantum repeater. The work represents significant progress in the infrastructure needed to communicate between dissimilar quantum states separated by tens of kilometres, and from there, one can perform all kinds of operations in a metropolitan-scale distributed quantum computer. The advances made here are an important step towards a Quantum Internet.

MEET THE PRODUCT



ID281 Superconducting Nanowire

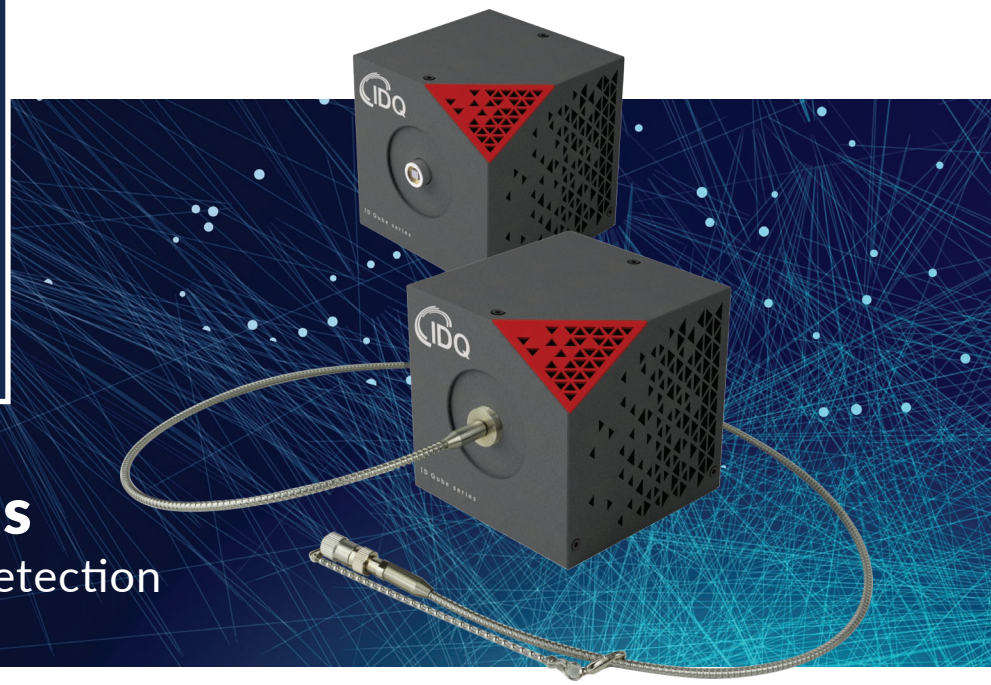
[1] 'Telecom-heralded entanglement between multimode solid-state quantum memories', D. Lago-Rivera, S. Grandi et al., 2021, Nature 594, 37–40 (2021)

[2] 'Entanglement between a telecom photon and an on-demand multimode solid-state quantum memory', J. V. Rakonjac, D. Lago-Rivera et al., 2021, arXiv:2106.05079

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Make every photon count



ID Qube Series

NIR Single Photon Detection

Synchronous and Asynchronous
Single-Photon Detection at Telecom Wavelengths

The ID Qube is a cost effective solution delivering precise and reliable single photon detection, with high detection efficiency, precise timing jitter and low detector noise. Available in two models:

- **The ID Qube NIR Free-Running** model, for applications in which asynchronous photon detection is essential, such as photon correlation or time of flight measurements.
- **The ID Qube NIR Gated** model, for applications in which synchronous photon detection is essential, such as quantum communication and QKD.

All models offer a gate input port connector, dedicated to avoiding saturation or undesired detections. The cooled InGaAs/InP avalanche photodiode and associated electronics have been designed to achieve especially low afterpulsing and dark count rates.

Get the best out of your photonic experiments and applications with the ID Qube NIR Series today.

KEY FEATURES

- **Compact, cost effective and dependable performance**
- **Fast gated** (up to 100 MHz) and **free-running**
- **Ultra-low noise** (<800 cps at 10%)
- **Low jitter** (<200 ps, typically <150 ps)
- **Free space or Fibre-coupled optical input**
- **Broadband detection** (900–1700 nm)

APPLICATIONS

- QKD and quantum communication
- Quantum optics and computing
- Single-photon source characterisation
- Fluorescence lifetime imaging
- Failure analysis of integrated circuits
- VIS, NIR and MIR spectroscopy

AVAILABLE IN EIGHT COMBINATIONS

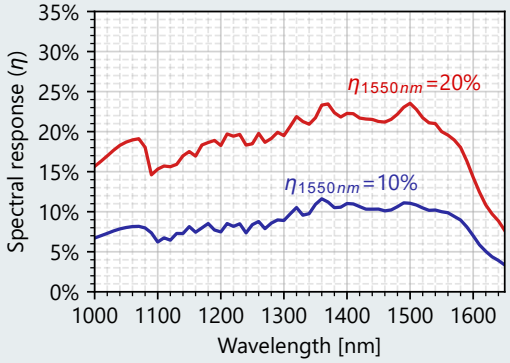
ID Qube-NIR-*XX*-*YY*-*ZZ*

- ***XX*** : **GAT** *(Gated model)* or **FR** *(Free running model)*
- ***YY*** : **FS** *(Freespace model)* or **MMF** *(Fibre coupled model, compatible with SMF and MMF FC/PC couplers)*
- ***ZZ*** : **STD** *(Standard noise model)* or **LN** *(Low noise model)*



Broadband performance

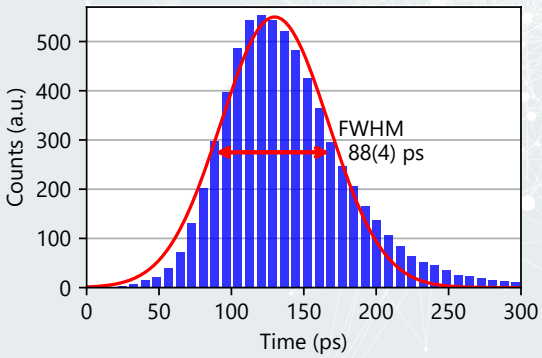
High system detection efficiency (SDE) across a broad range, characterized with carefully calibrated equipment.



(Above) Spectral response measurement for a typical ID Qube NIR device.

High precision

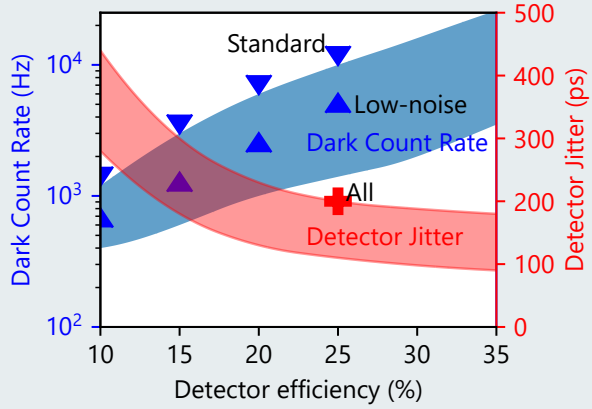
See the best timing resolution with the lowest detection timing jitter.



(Above) Timing jitter measurement of an ID Qube NIR detector at 30% SDE, including all other instrument jitter contributions, recorded with an ID1000 Time Controller.

Low noise

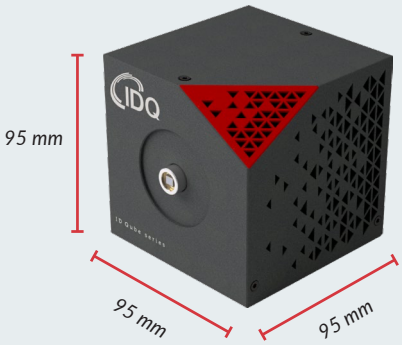
Always few dark counts with our Standard Qube models, and even better with the Low Noise model.



(Above) Dependence of detector noise and timing jitter with SDE. Users have the ability to balance the SDE, noise and precision to best fit their needs.

Compact

The ID Qube's small and compact form factor fits well into your experiments, and ideally suited for applications such as LiDAR, where compactness is key for system integration.

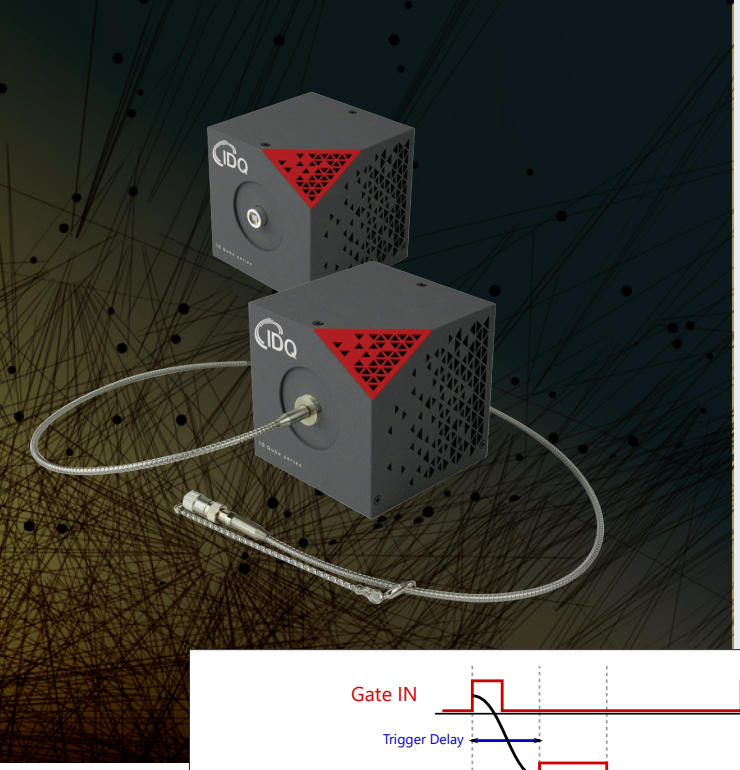


GATED AND FREE-RUNNING DETECTION AT TELECOM WAVELENGTHS

The ID Qube has been specially designed to achieve low dark count and afterpulsing rates.

The ID Qube NIR can operate at six detection efficiency levels[1][2] of 10%, 15%, 20%, 25%, 30% and 35% with a deadtime between 100 ns and 80 µs. In gated mode it accepts gates as short as 3 ns (500 ns for the Freerunning model) with a maximum repetition frequency of 100 MHz (1 MHz for the Free-running model).

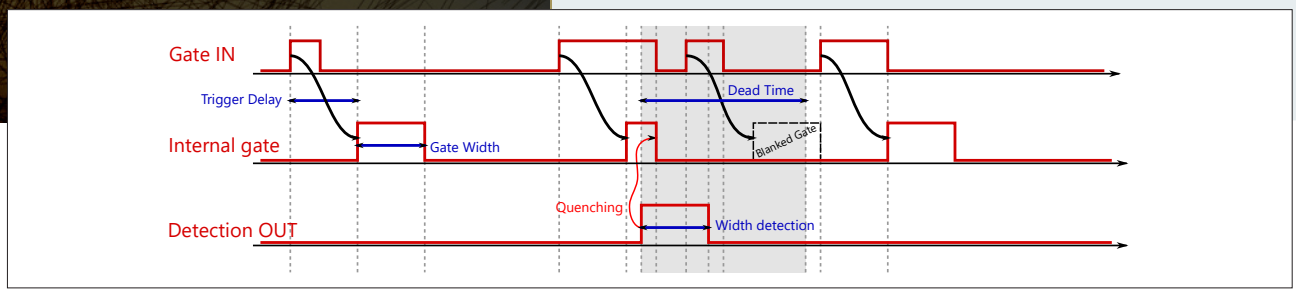
The arrival time of photons is reflected by a 10 ns LVTTL/NIM (user-selectable) pulse available at the SMA connector with a timing resolution typically as low as 150 ps at 25% efficiency. A simple USB interface allows the user to set the efficiency level and the deadtime.



GATED OPERATION

Detector gating — fast switching of the detector response — is achieved through an external electrical pulse source, such as an ID1000 Time Controller. Both models of the ID Qube NIR can be gated, with slow (maximum 1 MHz) gating of the ID Qube Free-running vs the fast (maximum 100 MHz) gating with the ID Qube Gated model.

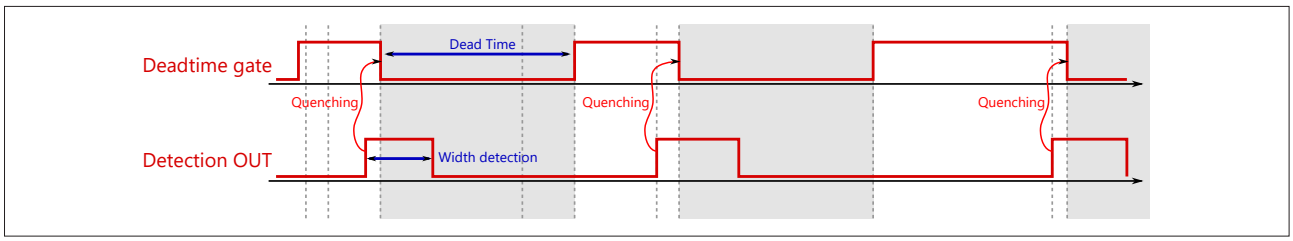
The ID Qube NIR Gated allows for high-speed synchronisation with external signals, such as in a fluorescence microscopy setup, while the slower gating of the ID Qube Free-running still proves useful for mitigating background noise.



(Above) Detector operation in Gated mode. Outside of a gate pulse, the ID Qube is insensitive to photon arrivals. In the case of small gate widths, the Gate IN signal can be used to trigger an internal adjustable-width gate. Whichever mode chosen, a photon-induced avalanche event inside a gate generates an electrical pulse, sent to the Detection OUT connector as a user's photon detection event signal. The quenching electronics then close the gate, halting the avalanche, while a user-adjustable deadtime is applied.

FREE-RUNNING OPERATION

All ID Qube NIR models can also be operated in free-running (asynchronous) mode. The APD is biased above its breakdown voltage in the so-called Geiger mode. Upon a photon absorption of a photon and the associated avalanche, the photon arrival time is reflected by the rising edge of a 10 ns-wide LVTTL/NIM pulse at the Detection OUT connector.



Both ID Qube Gated and ID Qube Free-running models are designed for fast avalanche quenching, strongly limiting after-pulsing (where additional 'false' detections are triggered by the tail of a 'real' detection pulse). This allows for even shorter deadtimes, increasing the maximum count rate without sacrificing performance.

TIME CONTROLLER SERIES BUNDLE

Take your experiment to the next level. Use the ID1000 Time Controller to register single-photon pulses and control the gates of up to five ID Qube detectors, within a combined time-tagger, pulse generator, delay generator package. All with advanced on-board logic for real-time four-fold coincidence measurements.



SPECIFICATIONS

ID Qube NIR Series				
Wavelength range	900 nm to 1700 nm			
Deadtime range	100 ns to 80 µs, in 100 ns steps			
Output pulse format	LVTTL or NIM			
Output pulse width	10 ns			
Optical coupling	Free space or optical fibre (MMF62.5)			
Efficiency range [1] calibrated at λ = 1550 nm	10%, 15%, 20%, 25%			
Extended efficiency range [2]	30%, 35%			
Timing jitter @ 25% efficiency level	Maximum 200 ps (150 ps typical)			
Noise performance @ efficiency level [3]	10%	15%	20%	25%
STD model (Max. dark count rate)	1.2 kHz	3 kHz	6 kHz	10 kHz
LN model (Max. dark count rate)	0.8 kHz	1.5 kHz	3 kHz	6 kHz
Gate-in max frequency	100 MHz (Gated model) / 1 MHz (Free-running model)			
Gate-in min pulse duration	3 ns (Gated model) / 500 ns (Free-running model)			
Gate-in voltage range	-2 V to 3 V			
Gate-in coupling	50 Ω DC			
Gate-in threshold voltage range	-2 V to 2 V, in 1 mV steps			
Output connector	SMA			
Operating temperature	+10°C to +35°C, max. 60% humidity			
Dimensions (W x H x L)	95 mm x 95 mm x 95 mm			
Weight	1 kg			
Cooling time @ power-on	< 3 minutes			
Power supply	100-240 VAC ; 1.4 A ; 50-60 Hz			
Storage temperature	+5°C to +50°C, max. 60% humidity			

Supplied Accessories:

- +12V, 60 W, AC/DC power adapter, with AC power cord
- Region adapted power cord
- 1.8 m USB cable
- Optical fibre cleaner (fibre-coupled model)
- C-mount adapter (free space model)
- Optical table mechanical adapter (M4 taps)
- 4 x Adhesive rubber feet

Notes:

(1) Additional efficiency levels can be calibrated on demand.

(2) The extended detection range is provided without guarantees of the device's noise performance. Above 25% efficiency, ID Qube devices start exhibiting non negligible afterpulsing, and detector dark counts can rise significantly. However, detector timing jitter has also been observed to improve with increasing detection efficiency.

(3) Dark count rate measured in free running mode with a 50 µs deadtime.

Applicable Standards

Safety US: UL 61010-1:2012 (3rd Ed.), AMD1
Safety CAN: C22.2 No 61010-1:2012 + U1:2015 + U2:2016 AMD1:2018

CE **Safety:** EN 61010-1:2010, AMD1:2016
EMC: EN 55032:2015, EN 55035:2017, EN 61326-1:2013

FC Title 47 Part 15:2019 Subpart B as a Class B device

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Make every photon count



ID1000
Time Controller Series

All in one time tagging, coincidence correlation and delay/pulse generation

Observe detection rates for up to five single-photon detectors, generate up to four coincidence histograms between any pair of connected detectors or trigger signals, and record every time tag for an exceptionally high number of detected photons, all with picosecond precision.

Beyond measurement, access four outputs with customizable and conditional pulses, to control and react to your experiments and applications in real time.

Get the most out of your photonic experiments and applications with the ID1000 Time Controller Series today.

KEY FEATURES

- **Five interchangeable input and reference channels**
  - High-resolution mode: up to 300 MHz across all inputs, with 1 ps resolution
  - High speed mode: up to 100 MHz per input, with 100 ps resolution
- **Measure four coincidence histograms in parallel**
- **Record the precise arrival time of every detection event,** with or without coincidence filtering
- **Tailored pulses (NIM or TTL) from four output channels,** with customizable patterns and widths
- **Advanced integrated logic:** filter events for two-, three- and four-fold coincidences in real time
- **Quick and easy lab integration:** suite of LabView and Python scripts

APPLICATIONS

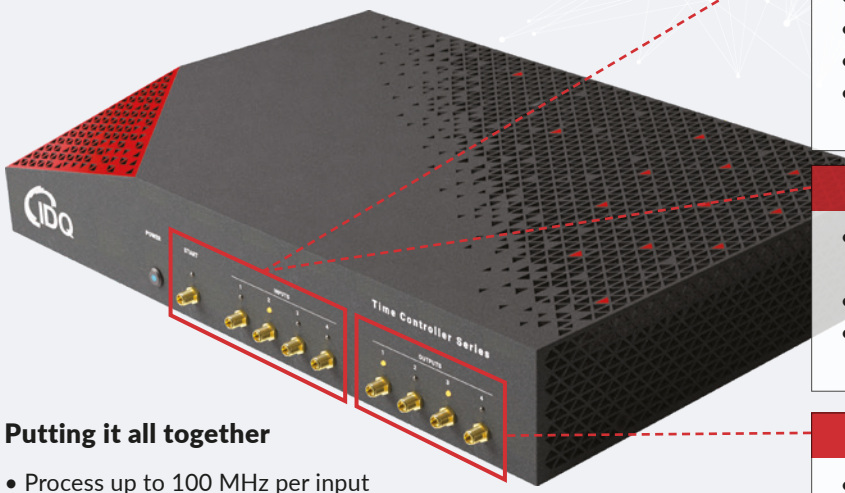
- QKD and quantum communication
- Quantum optics and computing
- Single-photon source characterisation
- Fluorescence lifetime imaging
- Failure analysis of integrated circuits
- VIS, NIR and MIR spectroscopy

NEW

- Measure more: multi-device synchronisation for over 64 input channels
- Enhanced precision: 1 ps resolution with < 4 ps rms jitter, and excellent DNL for low signal applications



ID1000 Time Controller Series



Counting and histograms

- Count single channel input rates
- Count 2/3/4-channel correlation rates
- Record four histograms simultaneously
- Up to 16,384 time bins, each between 1 ps and 4 ms wide

Time tagging

- Record every detection event, up to four timestamp measurements simultaneously
- Realtime 2/3/4-coincidence filtering
- Stream timestamps (raw or filtered) up to 10 M events/s

Output/delay generation

- Up to 4 x 250 MHz output pulses
- Customizable periodic or conditional on demand pulse sequences
- Digital NIM or TTL pulses, ideal for experimental instrument control

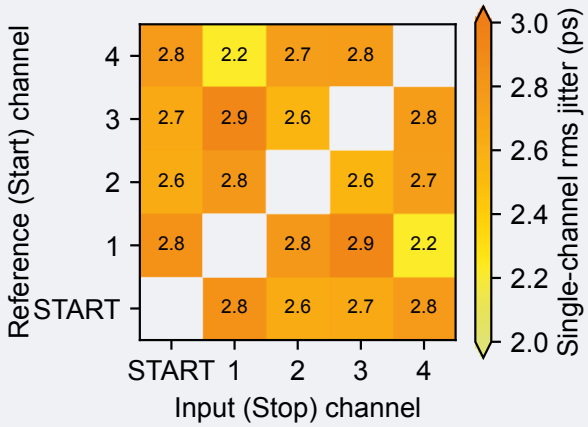
Putting it all together

- Process up to 100 MHz per input
- < 4 ps rms jitter (single-channel, HR)
- 1 ps digital resolution (HR)
- Five interchangeable input and reference channels
- Four programmable outputs

The last word in precision

Thanks to the advanced discrimination electronics of the ID1000 Time Controller, the precision and throughput of device will be future-proof for your lab, all in a versatile and cost-effective solution.

(Right) Timing jitter characterization of an example ID1000 Time Controller Series device in high resolution mode. Measured as the root-mean-squared (rms) width of a start-stop histogram between an Input channel and Reference channel pair, with fast synchronized pulses, divided by a factor of √2.



How to: TCSPC and the Instrument Response Function

The Instrument Response Function (IRF) describes the temporal response of your detection system against the 'true' dynamics of the system under investigation. The IRF takes into account all the detectors, measurement electronics, source precision, and any other instruments contributing to a non-instantaneous response in your experiments. Ultimately, the IRF tells you the best achievable timing precision of your detection system.

In the example to the right—typical for time-correlated single-photon counting (TCSPC)—the overall system response, $R_{Sys}(\tau)$, is a linear combination of the component IRFs, where $\otimes$ is the convolution operator.

The resulting uncertainty — the timing jitter $\Delta\tau_{Sys}$—is thus the quadrature sum of all component timing jitters. Use this insight to find the best ways to improve your experimental precision.



$$R_{Sys}(\tau) = R_{Src}(\tau) \otimes R_{Det}(\tau) \otimes R_{TC}(\tau) \otimes R_{Rest}(\tau)$$

$$(\Delta\tau_{Sys})^2 = (\Delta\tau_{Src})^2 + (\Delta\tau_{Det})^2 + (\Delta\tau_{TC})^2 + (\Delta\tau_{Rest})^2$$

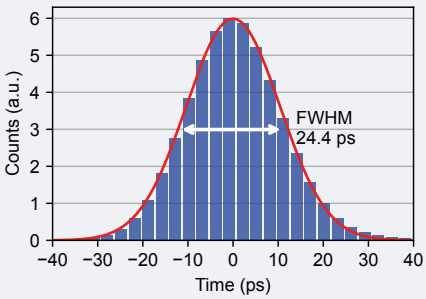
(Above) τ: time delay between detection events, 'Src': photon source, 'Det': single-photon detector pair, 'TC': time correlator (e.g. ID1000), 'Rest': everything else in the detection system contributing to a finite timing response.

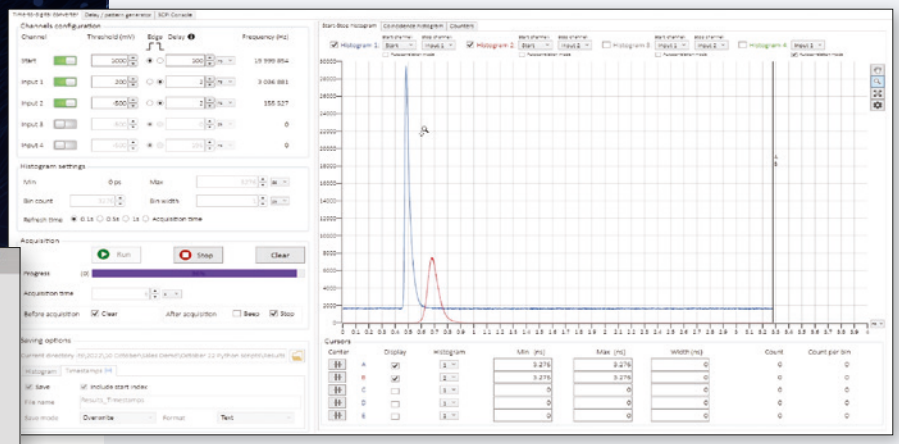
ID1000: made for you

Simpler than ever to use and integrate into your laboratory, **ID1000 Time Controller Series** devices come with a new-and-improved suite of LabView virtual instruments and Python scripts.

ID1000 Time Controllers are available in six combinations depending on your needs (see Specifications on the next page), each upgradeable at any time.

(Right) Example instrument response function of an ID281 SNSPD, at 1550 nm, recorded with an ID1000 Time Controller. The FWHM timing jitter of 24.4 ps includes all instrument jitter contributions, such that the real FWHM of the SNSPD is less than 22.3 ps.





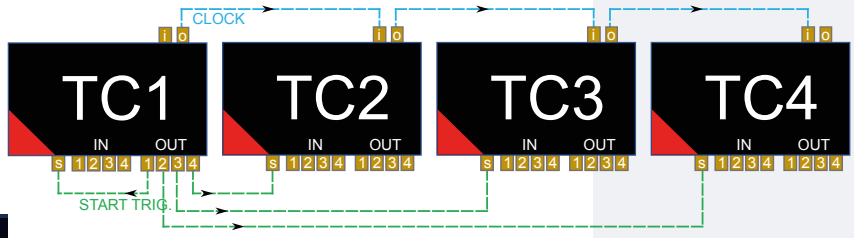


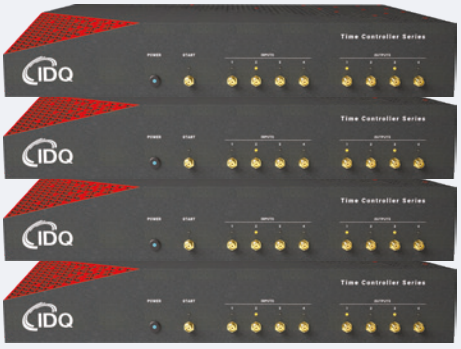
(Above) Control, operate and display measurements of your ID1000 Time Controller with the included GUI application.

(Left) Directly access and intuitively configure the ID1000 Time Controller's internal logic, with the included Configuration Editor application, for limitless customization.

In good company

- Operate multiple ID1000 Time Controller devices in unison
- Up to >64 synchronized input channels



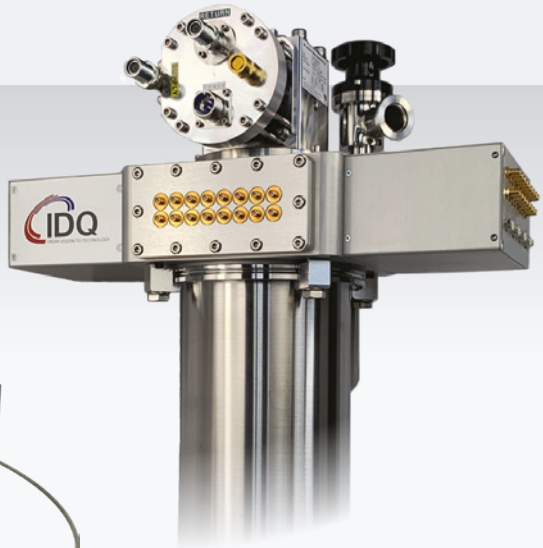


(Above) Example of four ID1000 Time Controllers in a multi-device configuration, giving access to 16 interchangeable input and reference channels.

You'll need some photons

The ID1000 Time Controller Series is designed especially for IDQ's range of single-photon detection solutions, from our versatile and easy-to-use semiconductor SPADs, to the game-changing performance of our SNSPDs.





SPECIFICATIONS

Functional specifications	High Speed	High Resolution[1]	MASTER	TCSPC
# Inputs channels per device	5		✓	✓
# Input channels in multi-device mode [2]	> 64		✓	
Min. time bin width (digital resolution)	100 ps	1 ps	✓	✓
Timing jitter (single-channel, rms) [3]	< 28 ps	< 4 ps	✓	✓
Max. DNL (rms)	< 10 ps	< 0.2 ps	✓	✓
Max. input count rate [4]	5 x 100 MHz	300 MHz	✓	✓
Max. timestamp rate [5]	10 M events/s		✓	✓
Input voltage range	-2 V to +2 V		✓	✓
Input channel delay	1 ps to 4 ms (in 1 ps steps)		✓	✓
# Output channels	4		✓	
Output pulse format	NIM or TTL		✓	
Max. rate per output channel	250 MHz		✓	
Min. output pulse width	6 ns (TTL), 1 ns (NIM)		✓	
Input — Output latency	400 ns	1500 ns	✓	
Electrical & Environmental specifications				
Power supply requirements	100 V to 240 V, 50 Hz to 60 Hz, 1 A to 2.5 A			
Operational temperature	5°C to 35°C			
Max. environment humidity	80% up to 31°C, 50% up to 35°C			
Device dimensions (W x H x L)	380 mm x 63 mm x 255 mm			

(1) HR add on license required.

(2) Number of accessible input channels when 16 Time Controllers are synchronised

(3) Measured as the width of a two channel start stop histogram with fast input pulses, divided by a factor of √2 to give the single channel jitter. Timing jitter for high speed mode is a worst case estimate with an assumed Gaussian instrument response profile.

(4) Count rates available for single channel counters, coincidence counters, and histogram processing, across all input channels of a single device.

(5) Measure up to four sets of timestamps (detection event time tags) in parallel, in total up to the max. timestamp rate.

On request, the two core ID1000 versions can be remotely upgraded with these add-on licenses:

Available add-ons	Core ID1000 version	
	MASTER	TCSPC
HR - Access to 'high resolution' (1 ps resolution, < 4 ps jitter) mode for the 5 input channels	⬇	⬇
PRCSG - Access to the internal FPGA logic, to configure real time selection filters for up to four fold coincidences	✓	⬇
4OUT - Access to the 4 output channels and the internal delay / pulse generator functionality	✓	⬇

⬇ Upgrade available ✓ already included

IDQ

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